

SW DIGITAL HUB

Frontend Developer Guide (FDG)

Version: 1.0

Project Type: Corporate Lead Generation Website

Framework: React.js + Vite + TypeScript

Styling: Tailwind CSS

1. Project Overview

Objective

Develop a premium, responsive, SEO-friendly corporate website for SW Digital Hub that showcases the company's services, portfolio, expertise, and generates qualified business enquiries.

The website should focus on:

- Brand Building
 - Lead Generation
 - Service Showcase
 - Portfolio Display
 - Performance
 - SEO
 - Excellent User Experience
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2. Technology Stack

Technology	Purpose
React 19	Frontend Library
Vite	Build Tool
TypeScript	Type Safety
Tailwind CSS	Styling
React Router DOM	Routing
Axios	API Requests
React Hook Form	Forms
Zod	Validation
Framer Motion	Animations
React Helmet Async	SEO
Swiper.js	Sliders
Lucide React	Icons

3. Project Folder Structure

sw-digital-hub/

public/

- favicon.ico
- robots.txt
- sitemap.xml
- manifest.json
- images/
- icons/
- videos/

src/

- app/
 - App.tsx
 - main.tsx
 - router.tsx
 - providers.tsx

- assets/
 - images/
 - videos/
 - icons/
 - fonts/

- components/
 - ui/
 - layout/
 - common/
 - forms/
 - sections/

- pages/

- services/
 - api.ts
 - contact.ts
 - audit.ts

- hooks/

- utils/

- constants/

- styles/

types/
config/

4. Website Pages

/

About

Services

Service Details

Portfolio

Case Studies

Blog

Blog Details

Testimonials

Contact

Free Digital Audit

Privacy Policy

Terms & Conditions

404 Page

5. Homepage Structure

Navbar

↓

Hero Banner

↓

Trusted By Clients

↓

Our Services



Why Choose SW Digital Hub



Our Process



Portfolio



Case Studies



Client Testimonials



FAQ



Free Digital Audit CTA



Contact Section



Footer

6. Components

Layout Components

Navbar

Footer

TopBar

Breadcrumb

ScrollToTop

Floating WhatsApp Button

Back To Top

Cookie Banner

UI Components

Button

Input

Textarea

Select

Badge

Card

Modal

Accordion

Tabs

Loader

Skeleton

Toast

Pagination

Dialog

Tooltip

Form Components

ContactForm

AuditForm

NewsletterForm

Section Components

Hero

Services

Process

Portfolio

Statistics

Testimonials

FAQ

CTA

Blog

Clients

Team

7. Routing

/

/about

/services

/services/:slug

/portfolio

/case-studies

/blog

/blog/:slug

/contact

/free-audit

/privacy-policy

/terms

/*

8. Responsive Breakpoints

Device	Width
Mobile	320px–639px
Tablet	640px–1023px
Laptop	1024px–1279px
Desktop	1280px–1535px
Large Desktop	1536px+

Use a **mobile-first** approach.

9. Design System

Primary Color

Company Blue

Secondary Color

Dark Gray

Accent Color

CTA Orange

Border Radius

- Small: 6px
- Medium: 12px
- Large: 20px

Shadow

- Card Shadow
 - Hover Shadow
 - Modal Shadow
-

10. Typography

Element	Style
H1	Hero Heading
H2	Section Title
H3	Card Title
Body	Paragraph
Caption	Small Text

Maintain consistent spacing and hierarchy across all pages.

11. Images

- Use **WebP** wherever possible.
 - Compress images before uploading.
 - Implement lazy loading for non-critical images.
 - Provide descriptive alt text for accessibility.
 - Avoid images larger than **250 KB** unless necessary.
-

12. Icons

Use **Lucide React** only.

Do not mix multiple icon libraries.

13. Animations

Use **Framer Motion** for:

- Hero entrance
- Section reveal
- Hover effects
- Counter animations
- Fade-in cards
- Button interactions

Avoid excessive or distracting animations.

14. Forms

Forms to implement:

- Contact Form
- Free Audit Form
- Newsletter Form

Use **React Hook Form** with **Zod** validation.

Validation rules:

- Name: Required
- Email: Valid email
- Phone: 10 digits
- Message: Required

Display inline validation errors and success/error messages after submission.

15. API Integration

Only two endpoints are required:

POST /api/contact

POST /api/free-audit

Create a centralized Axios instance and keep API logic inside the services folder.

16. SEO

Each page must include:

- Unique title
- Meta description
- Open Graph tags
- Canonical URL
- Structured data where applicable

Generate robots.txt and sitemap.xml.

17. Performance

- Lazy-load route components.

- Split code using dynamic imports.
 - Optimize images.
 - Use React.memo where appropriate.
 - Minimize unnecessary re-renders.
 - Keep Lighthouse Performance score above **90**.
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18. Accessibility

- Semantic HTML
- Keyboard navigation
- Visible focus states
- Proper labels for all form controls
- Meaningful alt attributes
- ARIA attributes where needed

Aim for **WCAG 2.1 AA** compliance.

19. Coding Standards

- Use **TypeScript** for all components.
 - Prefer functional components.
 - One component per file.
 - Keep components focused on a single responsibility.
 - Extract repeated logic into custom hooks.
 - Avoid inline styles.
 - Use descriptive names (ServiceCard, ContactForm, HeroSection).
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20. Git Workflow

main

↓

develop

↓

feature/homepage

↓

feature/contact-form

↓

feature/portfolio



feature/blog

Never commit directly to main.

21. Commit Convention

feat: add portfolio slider

fix: correct form validation

style: update hero spacing

refactor: simplify navbar logic

docs: update README

chore: upgrade dependencies

22. Testing Checklist

Before every merge:

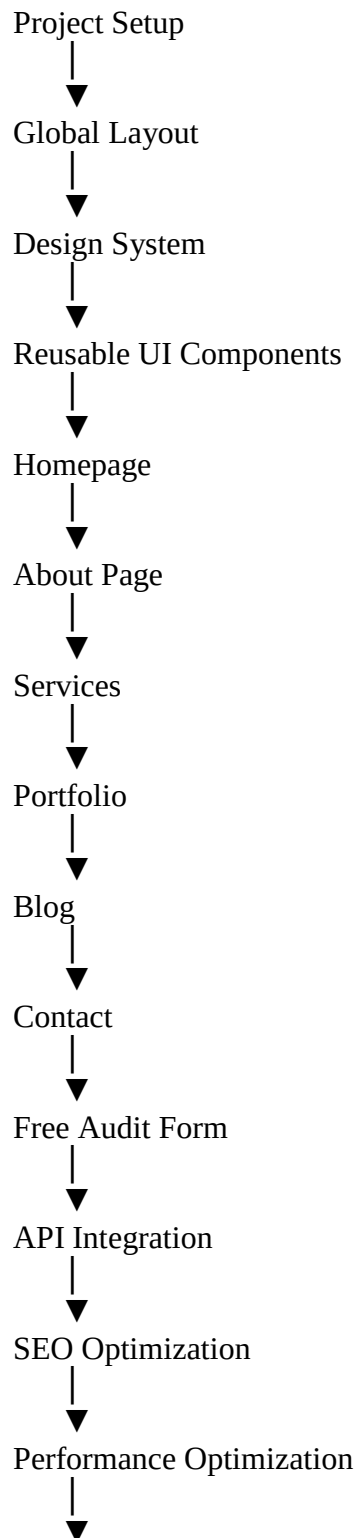
- No TypeScript errors
 - No ESLint warnings
 - Responsive on all supported devices
 - Forms validated
 - API calls tested
 - Images optimized
 - Accessibility checked
 - Lighthouse score ≥ 90
 - Cross-browser testing completed
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23. Production Checklist

- Environment variables configured
- Build succeeds without errors
- Sitemap generated
- Robots file configured
- Favicon added
- Metadata verified

- Analytics integrated
- Contact and Audit forms working
- 404 page configured

24. Development Roadmap



Testing



Production Deployment

Expected Deliverables

- Fully responsive React.js website
- Reusable component library
- SEO-ready pages
- Working Contact and Free Audit forms
- Clean TypeScript codebase
- Optimized assets
- High Lighthouse scores
- Production-ready build
- Developer documentation and README